Unit 6.1: Technical Proposal Writing

Siobhan Doherty

Post University: Computer Information Systems

CIS311: Technical Writing

Professors Franklin Orellana and Louis Unger

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Background

This proposal recommends replacing our current incandescent and fluorescent exit signs with self-illuminating exit signs. Doing so would lower energy costs, reduce maintenance work, and improve safety during power outages. Right now, our facility spends a growing amount of money on electricity, bulb replacements, and labor to keep traditional exit signs working. Similar problems were addressed in a project completed at the Marine Corps Development and Education Center (MCDEC) in Quantico, where switching to self-illuminating exit signs resulted in significant cost savings and a short return on investment.

Proposed Solution

Self-illuminating exit signs do not use electricity, wiring, or bulbs. Rather, they produce their own light for the complete life of the sign, which is usually 10 to 12 years. Because they do not rely on building power or backup batteries, these signs stay lit during power failures and emergencies. This makes them more dependable than traditional exit signs and helps ensure compliance with safety regulations.

Cost Analysis

The monetary benefits of this upgrade are apparent. Standard exit signs are always on and consume energy 24 hours a day. Fluorescent exit signs use roughly 13 to 26 watts, while incandescent signs use even more, between 50 and 100 watts. Studies have shown that upgrading exit signage can reduce energy and maintenance costs by over 90 percent because exit signs operate continuously, and inefficient models waste energy (Sardinsky & Hawthorne, 1994). At MCDEC Quantico, the switch resulted in first-year savings of about $37,000 at a total project cost of $97,000, leading to a payback period of just 2.6 years. Based on our current energy rates and the number of exit signs, we can expect similar long-term savings.

Maintenance savings are another primary benefit. Traditional exit signs require frequent bulb replacements, sometimes every few weeks. This adds labor, material, storage, and staff time costs. The Quantico project estimated savings of over $13,500 annually from eliminating bulb replacements and related labor. Self-illuminating exit signs remove these costs entirely, making expenses more predictable and easier to manage.

Installing self-illuminating exit signs is straightforward and causes very little disturbance. Installation is similar to hanging a picture, with an average cost of about $10 per sign. In existing buildings, unused electrical circuits can be turned off or used elsewhere. In new construction or renovations, these signs eliminate the need for dedicated electrical wiring, further reducing construction costs.

The initial purchase price of self-illuminating exit signs is competitive when electrical and labor costs are assessed. Single-face signs typically cost between $200 and $250, while double-face signs cost between $450 and $500. Bulk purchasing and contract pricing can additionally lower these costs. Over the lifetime of the signs, the total cost is greatly less than that of traditional exit signs.

Conclusion

In summary, replacing our existing exit signs with self-illuminating exit signs is a savvy and responsible choice. This upgrade will lower energy use, eliminate many maintenance needs, improve reliability during emergencies, and save money over time. Approving this proposal will improve safety while helping our organization better control operating expenses.

Reference:

Sardinsky, R., & Hawthorne, S. (1994). LED exit signs: Improved technology leads the way to energy savings (Publication No. TU-94-2). U.S. Department of Energy. https://www.osti.gov/biblio/55521